

ETHICAL AI REVIEW PACKAGE

Patient Triage Recommendation Model

INCOMPLETE — EVIDENCE-LIMITED DRAFT

How to read this document

Every entry below is either supported by a named source artifact or left blank. Nothing has been inferred, assumed, reconciled, or carried across from earlier draft answers.

Convention	Meaning
Normal text	Supported by the supplied evidence. The artifact it comes from is named in the Source column or note beneath.
[NOT SUPPLIED]	Cannot be completed from the materials provided. The reason is stated alongside.
CONFLICT	Two supplied artifacts state incompatible facts. Neither is treated as authoritative, so no value is recorded.

Source register

#	Artifact	Status
S1	Use-Case Intake Form	Used. Two fields left empty by the submitting team.
S2	Bias-Scan Report	Used. Undated and unsigned.
S3	Lineage Metadata Records (catalog screen)	Used. Conflicts with S4.
S4	Lineage Metadata Records (dashboard panel)	Used. Conflicts with S3 — different asset list and issues.
S5	Data Quality & Drift Dashboard	Used. No threshold definition accompanies it.
S6	Catalog asset view — vector-idx-cust-support-prod	Not usable. Customer-support asset, not this model.
S7	Bias Scan Report (dashboard panel)	Not usable. Conflicts with S2 on every value and attribute.

Executive Summary

[NOT SUPPLIED] *The summary must reflect findings from all four sections and close with a single governance decision. Sections 3 and 4 cannot be completed, and no section records a decision, so the summary has no body of evidence to reflect. Write this last, once the gaps below are closed.*

Governance Metadata

Field	Entry	Source
Model name	Patient Triage Recommendation Model	S1, S2
Model version	[NOT SUPPLIED]	—
Incident reference	[NOT SUPPLIED]	No source artifact carries an incident identifier
Review date	[NOT SUPPLIED]	No source artifact is dated
Deployment status	Live in production — Hospital / Emergency Department	S2
Intended purpose	Clinical Triage Support	S1

Field	Entry	Source
Primary stakeholders	Emergency Room Staff	S1
Target population	Adult and Elderly Patients	S1
Reviewer	[NOT SUPPLIED]	—
Approving authority	[NOT SUPPLIED]	No approver role is defined anywhere in the materials

Section 1 — Responsible AI Scoring

Part A: Principle-level scoring

Scoring key: Pass — principle adequately addressed. Concern — potential issue requiring design changes or compensating controls. Critical Flag — significant risk.

#	Principle	Score	Rationale	Source
1	Fairness	Critical Flag	Disparate Impact Ratio 0.68 against a threshold of ≥ 0.80 and below the P1 critical floor of 0.70; False Negative Rate Disparity 0.13 against ≤ 0.05 ; Demographic Parity Difference 0.12 against ≤ 0.05 . All three measured for age group 45–60 against a 25–34 reference.	S2
2	Reliability and Safety	Critical Flag	The scan records potential harm as systematic deprioritisation and delayed clinical attention leading to severe medical deterioration, higher mortality rates, and delayed critical care in emergency settings.	S2
3	Privacy and Security	Concern	The intake form states a commitment to protect patient privacy and data security under Ethical Considerations but records no technical safeguards, and the Regulatory Compliance Requirements field was left empty.	S1
4	Inclusiveness	Concern	Target population is recorded as Adult and Elderly Patients, but the scan assessed only ages 45–60 against 25–34. No elderly cohort was tested, and no gender or ethnicity attribute was assessed.	S1, S2
5	Transparency	Concern	The intake form records no explainability approach and no description of how recommendations are presented to clinical staff.	S1
6	Accountability	Concern	The intake form names Emergency Room Staff as stakeholders but records no escalation route and no human-in-the-loop protocol. The Known Risks & Limitations field was left empty.	S1

Part B: Overall decision

[NOT SUPPLIED] No rule maps principle scores to Proceed, Mitigate, or Reject. The two definitions supplied point to different outcomes: the scoring key describes a Critical Flag as risk that "may not be addressable through standard mitigation", while the governance checklist reserves Reject for "irremediable" violations. Until one aggregation rule is published, two Critical Flags cannot be resolved to a decision.

Part C: Follow-up actions

#	Action Required	Acceptance Criteria	Owner	Due Date	Source
1	Temporarily pause the automated prioritisation function and revert the Emergency Department to manual clinician triage.	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	S2

#	Action Required	Acceptance Criteria	Owner	Due Date	Source
2	Implement adversarial retraining / clinical debiasing to correct the identified age-group triage disparities.	Disparate Impact Ratio ≥ 0.80 ; False Negative Rate Disparity ≤ 0.05 ; Demographic Parity Difference ≤ 0.05 .	[NOT SUPPLIED]	[NOT SUPPLIED]	S2
3	Conduct an urgent ethical review of model metrics with Hospital Compliance, Clinical Safety and IRB teams.	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	S2
4	Deliver weekly validation audit reports to Clinical Governance Stakeholders prior to re-releasing the triage automation features.	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	S2

Source: S2 — Required Follow-Up Actions. Acceptance criteria for action 2 are derived from the metric thresholds stated in the same report. No owners or due dates appear in any supplied artifact.

Section 2 — Bias Findings and Mitigation

Bias findings summary

Attribute	Cohort vs Reference	Metric	Value	Threshold	P1 Escalation Rationale
Age	45–60 vs 25–34	Disparate Impact Ratio	0.68	≥ 0.80	Value is strictly below the P1 critical floor of 0.70.
Age	45–60 vs 25–34	False Negative Rate Disparity	0.13	≤ 0.05	Disparity exceeds the maximum allowable P1 threshold variance of 0.10.
Age	45–60 vs 25–34	Demographic Parity Difference	0.12	≤ 0.05	Disparity exceeds the maximum allowable P1 threshold variance of 0.10.

Source: S2. Impacted demographic attribute recorded as Age Group (45–60 vs 25–34).

CONFLICT A second bias scan (S7) reports different attributes and values for the same model — Disparate Impact 0.65 for Gender/Female, Equalized Odds 0.27 for Age 70+, and False Positive Rate Disparity 0.18 for Ethnicity/Hispanic — with no thresholds stated. No value, cohort, or metric matches S2. The two cannot both describe this model, so only S2 is recorded above and the discrepancy is left open.

Most critical finding

[NOT SUPPLIED] All three findings are recorded at Priority 1 and the scan supplies no ranking between them. Nothing in the supplied evidence designates one as most critical.

Mitigation options comparison

[NOT SUPPLIED] Only one strategy is supplied — In-processing Adversarial Debiasing and Objective Clinical Metric Constraints. No alternative controls, and no ratings for any alternative, appear in any source artifact, so no comparison can be performed.

Preferred mitigation — justification across the four evaluation dimensions

Dimension	Justification	Source
Ethical Effectiveness	Directly addresses clinical triage bias by penalising discriminatory prioritisation patterns during model optimisation, reducing false-negative and demographic disparities.	S2
Operational Feasibility	Can be integrated into existing secure clinical ML pipelines using established Python debiasing packages, without rebuilding integrated EHR databases.	S2
Reversibility	Retrained model checkpoints are version-controlled, allowing rollback to the standard clinical baseline if prediction degradation or safety anomalies occur.	S2
Auditability	Retraining logs, performance validation logs and fairness metrics are captured, allowing review by the Institutional Review Board and Clinical Governance Committee.	S2

Monitoring requirement statement

Element	Content	Source
What is tracked	Disparate Impact Ratio, Demographic Parity, False Negative Rate Disparity	S2
Cadence	Weekly automated batch audits	S2
Responsible owner	Lead Clinical Data Scientist / Hospital AI Ethics & Triage Governance Lead	S2
Trigger threshold	[NOT SUPPLIED]	No breach value is stated
Escalation path	[NOT SUPPLIED]	No route or response owner is stated
Duration	[NOT SUPPLIED]	No end condition is stated

Section 3 — Dataset Ethics Review

Data ethics checklist

#	Criterion	Designation	Rationale	Source
1	Sourcing Transparency	Flag	Lineage records show the source system not documented for Patient Admissions Data, and source details not documented for the demographic dataset.	S3, S4
2	Consent and Licensing	Fail	The lineage record for the demographic dataset states "Consent records not found". No lawful basis for the use of these attributes in model training is documented.	S3
3	Representational Fairness	Fail	Three metrics breach their thresholds at Priority 1 for age group 45–60 against a 25–34 reference.	S2
4	Data Minimization	[NOT SUPPLIED]	No supplied artifact addresses field-level scope, pseudonymisation handling, or a data dictionary.	—
5	Provenance Documentation	Flag	Lineage is recorded as incomplete for Triage Risk Scores with the processing step unverified, and timestamp metadata is missing for the vitals dataset.	S3, S4

Overall decision

[NOT SUPPLIED] No rule maps Pass / Flag / Fail designations to the decision, and the instruction offers only Approve or Reject — which cannot accommodate the two Flag designations recorded above. A third value, or a mapping rule, is needed before this can be recorded.

Lineage metadata review

CONFLICT Two lineage registers were supplied. They list different assets, different source systems, and different issues for the one dataset they share. Both are reproduced; neither is treated as authoritative.

Register A — catalog screen (S3)

Dataset	Source System	Lineage Status	Issue	Remediation Path
Patient Admissions Data	SQL Server	Missing Source Info	Source system not documented; source connection not documented; data retention policy missing	[NOT SUPPLIED]
Triage Risk Scores	Azure Data Lake	Incomplete Lineage	Processing step unverified	[NOT SUPPLIED]
Demographic Data	CSV Upload	Missing Metadata	Consent records not found	[NOT SUPPLIED]

Register B — dashboard panel (S4)

Dataset	Source System	Lineage Status	Issue	Remediation Path
Patient_History.csv	EHR System	Verified	Source verified	—
Lab_Results.csv	Lab_Database	Complete	Full lineage tracked	—
Demographics_Data	External Source	Missing	Source details not documented	[NOT SUPPLIED]
Vitals_Log.csv	IoT Device	Incomplete	Timestamp metadata missing	[NOT SUPPLIED]

Source: No remediation path for any non-compliant source appears in any supplied artifact.

Section 4 — Data Classification and Quality

Classification and retention tagging record

Embedding Index	Environment	Source	Sensitivity Label	Retention	Lineage Status
[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]
[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]

[NOT SUPPLIED] Four separate gaps prevent this record being completed. (a) The only embedding index in the supplied evidence is vector-idx-cust-support-prod, sourced from Blob Storage (Customer Transcripts) with customer_id and transcript_text columns — a customer-support asset, not this model. (b) No second index is supplied. (c) No retention value appears in any artifact. (d) No sensitivity label taxonomy is published, so no tag can be identified as the correct one.

Data quality findings as supplied

Feature	Issue	Metric	Value	Status as Supplied	Threshold Band
Heart Rate	Value Drift	PSI	0.29	Needs Review	[NOT SUPPLIED]
Blood Pressure	Missing Values	% missing	5.4%	At Risk	[NOT SUPPLIED]
Symptom Code	Category Drift	CSI	0.18	Action Required	[NOT SUPPLIED]

Source: S5. Training data drift recorded at 23%, operational data drift at 17%, with no threshold for either.

NOTE The statuses supplied are not mutually consistent: Heart Rate carries the highest recorded value (PSI 0.29) and the mildest status, while Symptom Code carries a lower value (CSI 0.18) and the strongest. They are reproduced above as supplied rather than corrected.

Remediation tickets

[NOT SUPPLIED] No quality threshold is defined in any supplied artifact, so which features cross it cannot be determined, and no proposed remediation action is supplied for any feature. Both are required before a ticket can be raised.

Cross-Section Integration Check

Check	Result	Finding
Does the bias finding escalated in Section 2 connect to a risk identified in Section 1?	Not met	Section 2 designates no single escalated finding, because the scan supplies no ranking. There is nothing to connect.
Does the non-compliant data source flagged in Section 3 appear in the Section 4 record?	Not met	Section 3 flags datasets; the Section 4 record covers embedding indexes only, and cannot be completed. A flagged dataset has nowhere to appear.
Do all sections consistently reference the same scenario details and data assets?	Not met	Two bias scans, two lineage registers, and one embedding index belonging to a different scenario. Conflicts are recorded, not resolved.
Does the package close with a single, clearly supported governance decision?	Not met	No section records a decision, because no rule maps scores or designations to one. The executive summary therefore cannot be written.

What This Package Needs to Be Completed

#	Required	Unblocks
1	A rule mapping principle scores to Proceed / Mitigate / Reject	Section 1 Part B; the executive summary
2	A rule mapping Pass / Flag / Fail to the Section 3 decision, with a third value for Flag	Section 3 decision
3	A designated most critical bias finding, or a ranking rule	Section 2; Integration check 1
4	Resolution of the two conflicting bias scans to one set of values	Sections 1, 2, 3
5	Resolution of the two conflicting lineage registers to one asset list	Section 3; Integration check 2
6	Two embedding indexes belonging to this model, with sensitivity labels and retention values	Section 4 tagging record
7	A published sensitivity label taxonomy and retention schedule	Section 4 tagging record
8	A defined data quality threshold	Section 4 tickets
9	Proposed remediation actions for each flagged feature and each non-compliant source	Section 3; Section 4
10	Alternative mitigation controls with ratings	Section 2 options comparison
11	Monitoring trigger threshold, escalation path and duration	Section 2 monitoring statement
12	Owners and due dates for follow-up actions	Section 1 Part C
13	Incident reference, review date, reviewer and approving authority	Governance metadata; sign-off
14	Evidence on field-level scope and pseudonymisation handling	Section 3 criterion 4

Authorisation

Role	Name	Signature	Date
Preparer	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]
Approving authority	[NOT SUPPLIED]	[NOT SUPPLIED]	[NOT SUPPLIED]

Source: No reviewer name, approver role, or signature appears in any supplied artifact.